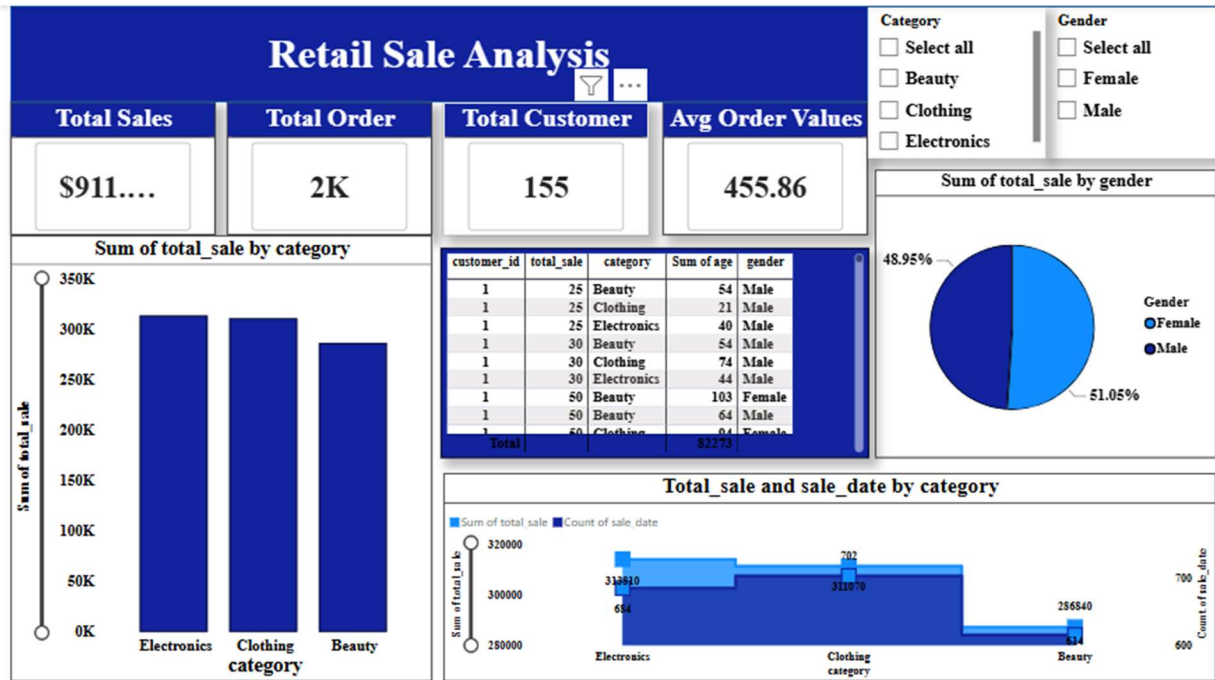


Retail Sales Analytics Case Study

Transforming Raw Excel Transactions into Business Intelligence Using SQL & Power BI



Business Context

Retail businesses generate large volumes of transactional data every day, including customer purchases, product categories, sales values, and purchasing behaviors. However, raw transactional data alone does not create business value. To support decision-making, data must first be structured, cleaned, analyzed, and transformed into meaningful insights.

In this project, I simulated the role of a Data Analyst working with a retail organization that wanted to better understand its sales performance, customer behavior, and revenue drivers.

The main business challenge was:

How can raw retail transaction data be transformed into reliable insights that improve sales strategy, customer understanding, and operational decision-making?

To solve this problem, I developed an end-to-end analytics workflow that transformed raw Excel transaction data into a structured SQL database, performed data cleaning and business analysis, and finally delivered insights through an interactive Power BI dashboard.

Phase 1: Understanding the Raw Business Data

The project started with a raw retail transaction dataset stored in Excel format. Before performing any analysis, the first step was to understand the structure, meaning, and business context of the data. The initial analysis focused on understanding what each transaction represented, how customers, products, and sales were connected, which fields were required for business reporting, and whether any data quality issues could impact analytical results.

The dataset contained transaction-level information, including transaction identifiers, customer information, product categories, purchase dates and times, quantity purchased, unit prices, cost information, and total sales values.

At this stage, the main objective was not only to calculate sales metrics, but to understand how operational business data is generated, organized, and prepared for analysis.

Phase 2: Building a Structured SQL Database

Although Excel is commonly used for storing and exploring business data, it becomes difficult to manage and scale when performing complex analysis on larger datasets.

To create a more reliable analytical environment, I migrated the retail transaction data into a relational MySQL database.

The database development process involved importing Excel/CSV transaction data into MySQL, designing structured database tables, defining appropriate data types, creating primary keys for transaction identification, and organizing transactional information into a structured analytical schema. This transformation converted a simple spreadsheet into a database environment that supports repeatable analysis, improved data management, and scalable reporting.

Through this phase, I developed practical experience in relational database design, SQL database creation, data modeling, and schema organization.

Phase 3: Data Cleaning and Quality Validation

Before generating business insights, the dataset needed to be accurate, consistent, and reliable. In real-world analytics projects, poor data quality can lead to incorrect conclusions and ineffective business decisions. Therefore, I performed SQL-based data validation and cleaning to improve the

reliability of the dataset. The cleaning process focused on identifying missing values in important business fields, including customer ID, product category, quantity, total sales, sale date, sale time, gender, and cost information.

I also checked for duplicate transactions to ensure that sales metrics were not affected by repeated records. Another important step was improving data consistency by standardizing categorical values. For example, different variations of the same category such as "clothing", "Clothing", and "CLOTHING" were standardized into a consistent format.

Additionally, invalid records such as zero or negative quantities and incomplete transactions were identified and removed or corrected.

The SQL techniques applied during this stage included filtering conditions using WHERE statements, missing value validation using IS NULL checks, DELETE operations, text cleaning using TRIM functions, and data standardization techniques. The result was a clean and reliable dataset ready for business analysis.

Phase 4: Business Analysis Using SQL

After preparing the database, SQL was used as an analytical tool to answer practical business questions. The objective was not only to calculate numbers, but to understand business performance, customer behavior, and sales patterns.

The analysis focused on three major business areas: product performance, customer value, and purchasing behavior.

Product Revenue Analysis

One of the key business questions was:

Which product categories generate the highest revenue and transaction volume?

To answer this question, I analyzed sales contribution across different product categories by calculating total revenue, number of transactions, and category performance.

SQL aggregation functions such as SUM(), COUNT(), and GROUP BY were used to generate category-level business metrics.

The results helped identify which categories contributed the most revenue and provided insights that could support product strategy, inventory planning, and sales optimization.

Customer Value Analysis

Another important business question was:

Who are the highest-value customers contributing the most revenue?

Customer-level analysis was performed by calculating total customer spending, transaction frequency, and revenue contribution.

SQL techniques including aggregation, GROUP BY, ORDER BY, and LIMIT were applied to identify the top-spending customers.

This analysis helped reveal valuable customer segments and provided insights that could support customer retention strategies and targeted marketing decisions.

Sales Timing Analysis

A further area of analysis focused on customer purchasing behavior.

The business question was:

When do customers purchase the most?

Sales patterns were analyzed based on month, year, and transaction time. Customer purchases were classified into different time periods, including morning, afternoon, and evening.

Date extraction functions, CASE statements, and time-based grouping were used to identify purchasing patterns.

These insights can support operational decisions such as employee scheduling, promotional timing, and demand planning.

Phase 5: Advanced SQL Analytics

To move beyond basic reporting, advanced SQL techniques were applied to generate deeper analytical insights. Customer segmentation was performed by creating age-based customer groups, including Young, Adult, Middle Age, and Senior segments.

Transaction values were also classified into different sales levels, including low-value, medium-value, and high-value transactions.

Window functions such as RANK() and DENSE_RANK() were used to identify top-performing transactions and category rankings.

Additionally, SUM() OVER() was applied to calculate cumulative sales trends over time and understand revenue progression.

These techniques demonstrated the ability to perform advanced SQL-based analytics and transform transactional data into business intelligence.

Phase 6: Power BI Business Intelligence Dashboard

After completing the SQL analysis, the cleaned and transformed dataset was connected to Power BI to create an interactive business intelligence dashboard.

The purpose of the dashboard was to convert analytical results into an easy-to-understand reporting solution for business stakeholders.

The dashboard provides visibility into overall sales performance, including total revenue, transaction volume, customer count, and revenue trends. It also provides sales analysis by showing category performance, monthly sales patterns, and revenue contribution.

Customer analytics features help understand purchasing behavior, customer segments, and high-value customers. Product insights highlight best-performing categories and sales distribution patterns.

The final dashboard transforms technical SQL analysis into clear business storytelling and supports data-driven decision-making.

Final Business Insights Delivered

Through this project, I was able to answer important business questions, including how much revenue the retail business generates, which product categories contribute the most sales, who the highest-value customers are, which customer groups generate more revenue, when customers purchase the most, which months perform best, which transactions have the highest business value, and how historical sales data can support future planning.

Project Outcome

This project represents a complete Data Analyst workflow, starting from raw Excel transactions and progressing through database development, data quality improvement, SQL-based analytics, and Power BI business intelligence reporting.

The complete process followed this analytical journey:

Raw Excel Transactions

→ Understanding and preparing business data

SQL Database Development

→ Creating a structured analytical environment

Data Cleaning and Validation

→ Improving accuracy and reliability

Business Analysis

→ Developing KPIs and answering business questions

Power BI Dashboard

→ Transforming analysis into visual insights

Decision Support

→ Delivering actionable business recommendations

Skills Demonstrated

Through this project, I developed practical skills in SQL database development, relational data modeling, data cleaning and validation, exploratory data analysis, business KPI development, customer analytics, sales performance analysis, advanced SQL window functions, Power BI dashboard development, and data storytelling.

SQL Retail Sales Analysis – Business Questions

1. Data Cleaning Questions

Data Quality Checks

1. Check for duplicate transactions

- Write a SQL query to identify duplicate records based on `transaction_id`.

2. Remove duplicate transactions

- Write a SQL query to remove duplicate records while keeping the first occurrence of each transaction.

3. Identify missing values

- Write a SQL query to identify rows with NULL values in important columns:
 - `customer_id`
 - `category`
 - `quantity`
 - `total_sale`
 - `sale_date`
 - `sale_time`

- gender
 - cogs
- 4. **Handle missing customer age values**
 - Write a SQL query to replace NULL values in the `age` column with the average customer age.
- 5. **Handle missing sales values**
 - Write a SQL query to replace NULL values in `total_sale` with 0.
- 6. **Standardize product categories**
 - Write a SQL query to standardize category values so that:
 - clothing
 - Clothing
 - CLOTHING

appear consistently as one category.
- 7. **Remove extra spaces**
 - Write a SQL query to trim unnecessary spaces from text columns such as:
 - category
 - gender
- 8. **Convert date format**
 - Write a SQL query to convert `sale_date` into a proper DATE format if stored as text.
- 9. **Identify invalid quantities**
 - Write a SQL query to identify transactions where:
 - $\text{quantity} \leq 0$
- 10. **Standardize gender values**
 - Write a SQL query to identify inconsistent gender values such as:
 - M
 - Male
 - male
 - F
 - Female

2. Exploratory Data Analysis (EDA) Questions

11. Count total transactions

- How many sales transactions exist in the dataset?

12. Count unique customers

- How many unique customers are present in the dataset?

13. Identify product categories

- What are the unique product categories available?

14. Understand dataset structure

- What are the main characteristics and distributions of:
 - customers
 - products
 - transactions
 - sales?

3. Business Analysis Questions

Sales Analysis

Q1. Daily Sales Analysis

Retrieve all columns for sales made on:

2022-11-05

Q2. Category and Quantity Analysis

Find all transactions where:

- Category = Clothing
- Quantity sold is greater than 4
- Month = November 2022

Q3. Revenue by Category

Calculate total sales for each product category.

Metrics:

- Total revenue
- Number of orders

Q4. Customer Demographic Analysis

Find the average age of customers who purchased products from the Beauty category.

Q5. High-Value Transactions

Find all transactions where total sales are greater than 1000.

Customer Analysis

Q6. Gender and Category Performance

Calculate the total number of transactions made by each gender within each category.

Analyze:

- Category
- Gender
- Transaction count

Q7. Monthly Sales Performance

Calculate average sales for each month and identify the best-performing month in each year.

Requirements:

- Extract year
- Extract month
- Calculate average sales
- Rank months by performance

Q8. Top Customers by Revenue

Find the top 5 customers based on highest total spending.

Metrics:

- Customer ID
- Total sales contribution

Q9. Customer Category Analysis

Find the number of unique customers who purchased products from each category.

Metrics:

- Category
- Unique customer count

Time-Based Analysis

Q10. Sales Shift Analysis

Create customer purchasing shifts based on transaction time.

Classify sales into:

- Morning:
 - Before 12 PM
- Afternoon:
 - Between 12 PM and 5 PM
- Evening:
 - After 5 PM

Calculate:

- Number of orders per shift

4. Advanced SQL Analysis Questions

Q11. Sales Classification

Classify transactions into sales groups using CASE statements.

Categories:

- Low Sales
- Medium Sales

- High Sales

Based on:

- total_sale value

Q12. Customer Age Segmentation

Create customer age groups using CASE statements.

Segments:

Age Range	Group
<25	Young
25–40	Adult
41–60	Middle Age
>60	Senior

Q13. Transaction Ranking

Rank transactions based on total sales within each category.

Required SQL:

- Window Function
- RANK()

Q14. Running Sales Total

Calculate cumulative sales over time.

Required SQL:

- SUM() OVER()
- Date ordering

Q15. Top Product Performance

Identify the top-selling product/category within each category using:

- DENSE_RANK()
- Window Functions

Final Business Questions Answered by This Project

The SQL analysis answers:

1. How much revenue does the business generate?
2. Which categories drive the most sales?
3. Who are the highest-value customers?
4. Which customer segments purchase more?
5. When do customers purchase the most?
6. Which months perform best?
7. Which transactions generate the highest revenue?
8. How can sales data support inventory and marketing decisions?